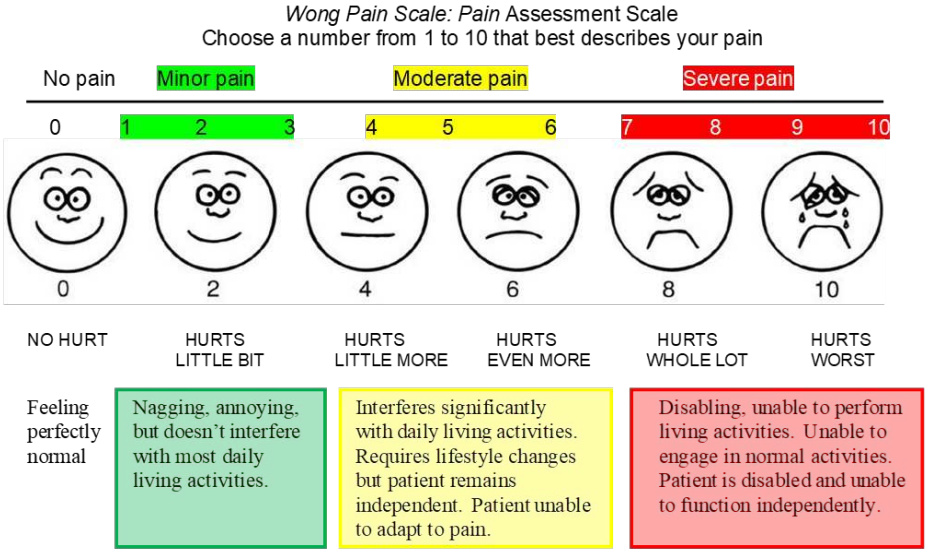
PAIN MANAGEMENT

For patients with suspected cardiac chest pain, refer to the **Chest Pain/Acute Coronary Syndrome Treatment Protocol**.

Patient care provided under this protocol is focused on reducing the level of pain for patients in the prehospital setting.

All pain should be assessed and scored using the "Wong Pain Scale." Reassessment should be timed according to medication onset of action, changes in patient condition, patient positioning, and other treatments. Pain treatment should be based on pain scale but may need modification based on patient assessment and/or condition.



Note: Medical Control contact is required for patients with special circumstances such as having an established care plan that deters pain management or having an established pain management care plan that differs in dose or administration from this protocol.

1. Place the patient in a position of comfort.
2. Verbally reassure the patient to control anxiety.
3. Administer basic interventions per applicable protocol (e.g., positioning, splinting, ice, etc.). If pain does not improve with basic interventions, consider analgesia.
4. Start an IV if required for medication administration, or per applicable treatment protocol. Refer to **Vascular Access & IV Flued Therapy Procedure Protocol**.
5. For pediatric patients (< 14 years), utilize MI MEDIC for appropriate medication dosage. When unavailable utilize pediatric dosing listed within protocol.
6. Per MCA selection, for mild to moderate pain (described as a 1-6 on the Wong Pain Scale), consider non-opioid analgesia.
7. For patients with suspected kidney stone pain at any level, **ketorolac** should be the first line medication if available.

MCA Selected Non-Opioid Analgesia

(MCA must select at least one)

☑ **Acetaminophen:**

1. Adults (patients > 14 years of age), administer 650 mg PO
2. Pediatrics refer to MI MEDIC. When MI MEDIC is unavailable refer to dosing table below.

☑ **Ibuprofen**

1. Adults (patients > 14 years of age), administer 400 mg.
    a. Do NOT use in pregnant patients.
2. Pediatrics (patients > 6 months of age and < 14 years of age), refer to MI MEDIC. When MI MEDIC is unavailable refer to dosing table below.

☑ **Ketorolac** (Toradol ®)

1. Adults (patients >14 years of age), administer 15 mg IM/IV. Do NOT use in pregnant patients.
2. Pediatrics (patients > 5 years of age and < 14 years of age refer to MI MEDIC. When MI MEDIC is unavailable administer 0.5 mg/kg IM/IV (max dose 15 mg)

Children's Elixir Dosing Table

Child's Weight	Child's Age	Acetaminophen 160 mg/5mL	Ibuprofen 100 mg/5mL
3-5 kg (6-12 lbs.)	0-2 mos.	1.25 mL (40 mg)	DO NOT GIVE
6-7 kg (13-16 lbs.)	3-6 mos.	3 mL (96 mg)	DO NOT GIVE
8-9 kg (17-20 lbs.)	7-10 mos.	4 mL (128 mg)	4 mL (80 mg)
10-11 kg (21-25 lbs.)	11-18 mos.	5 mL (160 mg)	5 mL (100 mg)
12-14 kg (26-31 lbs.)	19 mos.-35 mos.	6 mL (192 mg)	6 mL (120 mg)
15-18 kg (32-40 lbs.)	3-4 yrs.	7 mL (224 mg)	7.5 mL (150 mg)
19-23 kg (41-51 lbs.)	5-6 yrs.	9 mL (288 mg)	9.5 mL (190 mg)
24-29 kg (52-64 lbs.)	7-9 yrs.	12 mL (384 mg)	13 mL (260 mg)
30-36 kg (65-79 lbs.)	10-14 yrs.	15 mL (480 mg)	15 mL (300 mg)

1. For patients with severe pain (described as ≥ 7 on the Wong Pain Scale), consider either **ketamine** or opiate analgesia if applicable per MCA selection.
2. Patients shall receive only **ketamine** OR a single opioid medication. These may be used in conjunction with nonopioid options for optimal pain control.

MCA Selection for Ketamine use in pain management

☐ **Ketamine** not permitted.



☐ Contact Medical Control prior to **Ketamine** administration

☑ Administer **Ketamine**

MCA Selected Opioid Analgesia
(Must select at least one)

☐ **Morphine**

1. Adults (patients > 14 years of age), administer 0.1 mg/kg IV/IO/IM (maximum single dose 5 mg).
   a. IV/IO may repeat three times (total of four doses). Total dose may not exceed 20 mg. Interval between doses is minimally 10 minutes.
   b. If IM administration, may NOT repeat.
2. Pediatrics (patients > 18 months of age and < 14 years of age), refer to MI MEDIC. When MI MEDIC is unavailable, administer 0.1 mg/kg IV/IO/IM (maximum single dose 5 mg).
   a. IV/IO may repeat three times. Total dose may not exceed 20 mg. Interval between doses is minimally 10 minutes.
   b. If IM administration may NOT repeat.
3. Do NOT administer Morphine to children < 18 months of age.

☑ **Fentanyl**

1. Adults (patients > 14 years of age), administer 1 mcg/kg IV/IO/IM/IN dosage, rounded as per below:
   a. Up to and including 25 kg - administer 25 mcg
   b. 26-50 kg – administer 50 mcg
   c. 51-75 kg - administer 75 mcg
   d. 76 kg and above – administer 100 mcg
   e. IV/IO/IN may repeat one-time, total dose may not exceed 200 mcg. Interval between doses is minimally 10 minutes.
   f. If IM administration may NOT repeat
2. Pediatrics (patients < 14 years of age), refer to MI MEDIC. When MI MEDIC is unavailable, administer 1 mcg/kg IV/IO/IM/IN
   a. May repeat IV/IO/IN one time. Interval between doses is minimally 10 minutes.
   b. If IM administration may NOT repeat

3. Administer opioids slowly when using IV or IO routes. Systolic BP should be maintained at >100 mmHG for adult patients and >80 + (2 x age) mmHg for pediatric patients.

4. **Ketamine** for pain management given IV/IO should be diluted in 100 ml normal saline and administered via slow infusion over 5-10 minutes to avoid dissociation symptoms.



5. **Ketamine** may be administered IV/IO/IN as listed below.
   a. Adults (patients > 14 years of age):
      i. 0.2 mg/kg IV/IO (diluted in 100 ml NS), maximum single dose 25 mg
      ii. 0.5 mg/kg IN, maximum single dose 50 mg
      iii. May be repeated one time, at least 10 minutes after initial dose



b. Pediatrics (>6 years of age and <14 years of age) refer to MI MEDIC. If MI MEDIC is unavailable:

i. 0.2 mg/kg IV/IO (diluted I 100 ml NS), maximum single dose 25 mg

ii. 0.5 mg/kg IN, maximum single dose 50 mg

iii. May be repeated one time, at least 10 minutes after initial dose

 c. Pediatrics (>6 months of age and <6 years of age) refer to MI MEDIC. If MI MEDIC is unavailable:

i. 0.5 mg/kg IN

ii. May repeat one time, at least 10 minutes after initial dose

6. If an IV is not available, a single dose may be given IM.

 7. Do not administer additional pain medications after IM administration without online medical direction.

 8. Patients that receive **ketamine** or opioid analgesia require additional monitoring and administration considerations.

a. Continuous pulse oximetry monitoring

b. IV/IO administration must be performed slowly

c. IM administrations are single dose. No additional doses or other pain management medications may be administered.

 9. If nausea develops, refer to **Nausea and Vomiting Treatment Protocol**.

 10. For patients with evidence of hypotension or hypoperfusion, contact medical control

11. For patients with refractory pain after **ketamine** or opiate administration, contact medical control to consider additional analgesic agent

Medication Protocols: Acetaminophen, Fentanyl, Ibuprofen, Ketamine, Ketorlac, Morphine